

Dietary intake of n–3 and n–6 fatty acids and the risk of clinical depression in women: a 10-y prospective follow-up study

Background: The associations between different sources of dietary n–3 (omega-3) and n–6 (omega-6) fatty acids and the risk of depression have not been prospectively studied. **Objective:** The objective was to examine the relation between different n–3 and n–6 types with clinical depression incidence. **Design:** We prospectively studied 54,632 US women from the Nurses' Health Study who were 50–77 y of age and free from depressive symptoms at baseline. Information on diet was obtained from validated food-frequency questionnaires. Clinical depression was defined as reporting both physician-diagnosed depression and regular antidepressant medication use. **Results:** During 10 y of follow-up (1996–2006), 2823 incident cases of depression were documented. Intake of long-chain n–3 fatty acids from fish was not associated with depression risk [relative risk (RR) for 0.3-g/d increment: 0.99; 95% CI: 0.88, 1.10], whereas α -linolenic acid (ALA) intake was inversely associated with depression risk (multivariate RR for 0.5-g/d increment: 0.82; 95% CI: 0.71, 0.94). The inverse association between ALA and depression was stronger in women with low linoleic acid (LA) intake (P for interaction = 0.02): a 0.5-g/d increment in ALA was inversely associated with depression in the first, second, and third LA quintiles [RR (95% CI): 0.57 (0.37, 0.87), 0.62 (0.41, 0.93), and 0.68 (0.47, 0.96), respectively] but not in the fourth and fifth quintiles. **Conclusions:** The results of this large longitudinal study do not support a protective effect of long-chain n–3 from fish on depression risk. Although these data support the hypothesis that higher ALA and lower LA intakes reduce depression risk, this relation warrants further investigation.